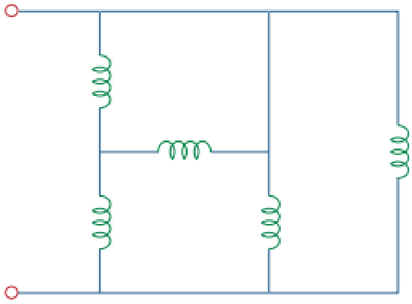
Problem

Find the equivalent inductance of the circuit in Fig. 6.72. Assume all inductors are 40 mH.

Fig. 6.72:



Step-by-step solution

Step 1 of 8

Refer to Figure 6.72 in the text book.

Draw the modified circuit diagram.

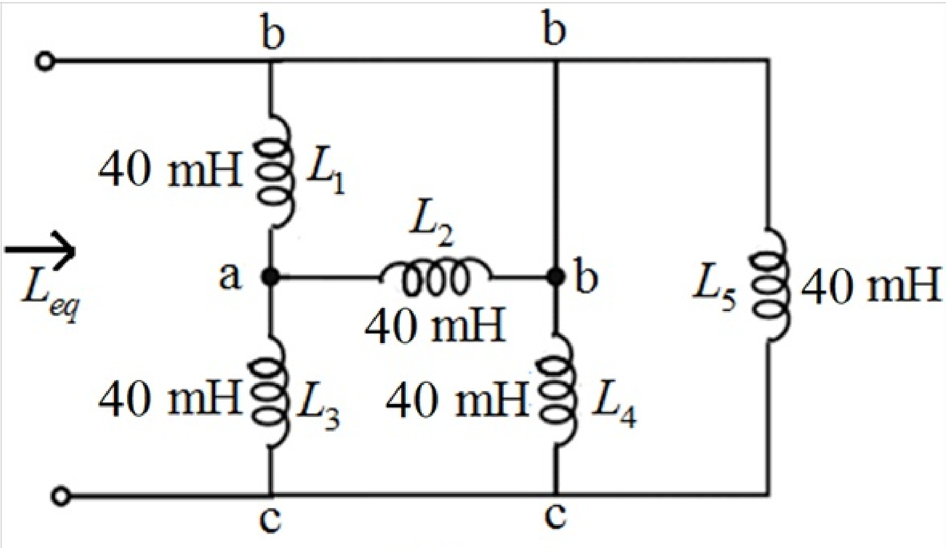


Figure 1

Step 2 of 8

In Figure 1, the inductors L_4 and L_5 are in parallel.

$$\begin{aligned} L_{eq1} &= (L_4 \parallel L_5) \\ &= (40\times10^{-3}) \parallel (40\times10^{-3}) \\ &= \frac{(40\times10^{-3})(40\times10^{-3})}{(40\times10^{-3})+(40\times10^{-3})} \\ &= 20\times10^{-3} \\ L_{eq1} &= 20 \text{ mH} \end{aligned}$$

Step 3 of 8

Draw the equivalent circuit diagram.

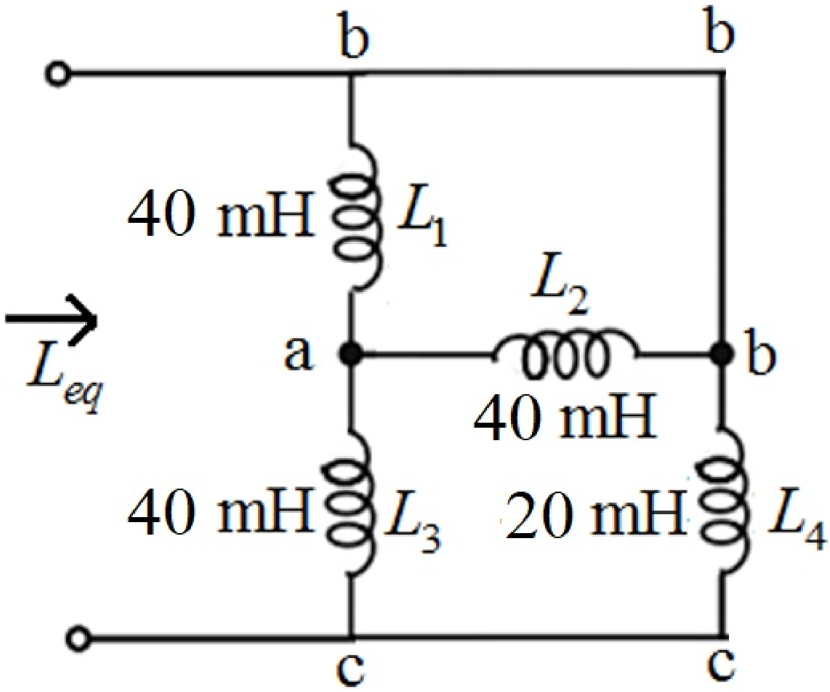


Figure 2

Step 4 of 8

In Figure 2, the inductors L_1 and L_2 are in parallel.

$$
\begin{aligned}
L_{eq2} &= (L_1 \parallel L_2) \\
&= (40\times10^{-3}) \parallel (40\times10^{-3}) \\
&= \frac{(40\times10^{-3})(40\times10^{-3})}{(40\times10^{-3})+(40\times10^{-3})} \\
&= 20\times10^{-3} \\
L_{eq2} &= 20 \text{ mH}
\end{aligned}
$$

Step 5 of 8

Draw the equivalent circuit diagram.

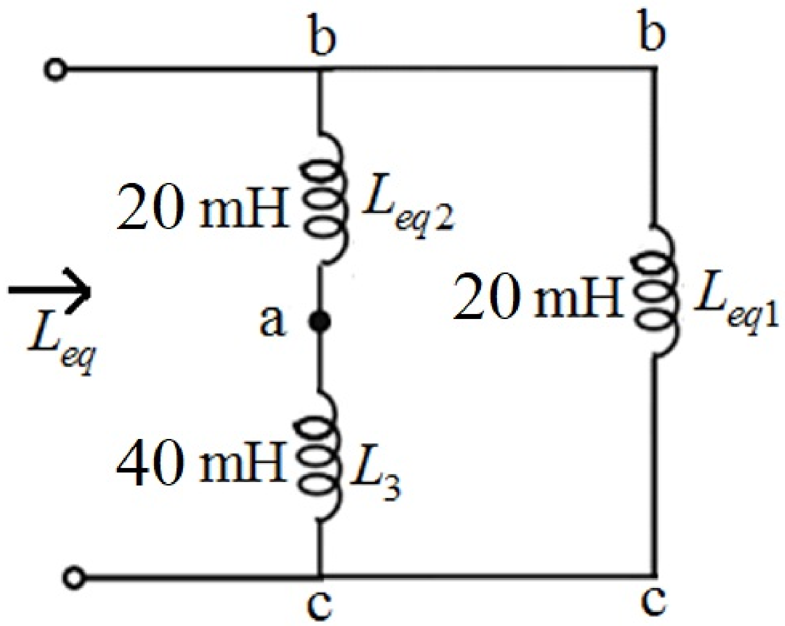


Figure 3

Step 6 of 8

In Figure 3, the inductors L_{eq2} and L_3 are in series.

$$\begin{aligned} L_{eq3} &= L_{eq2} + L_3 \\ &= \left(20\times10^{-3}\right)+\left(40\times10^{-3}\right) \\ &= 60\times10^{-3} \\ &= 60\text{ mH} \end{aligned}$$

Step 7 of 8

Draw the equivalent circuit diagram.

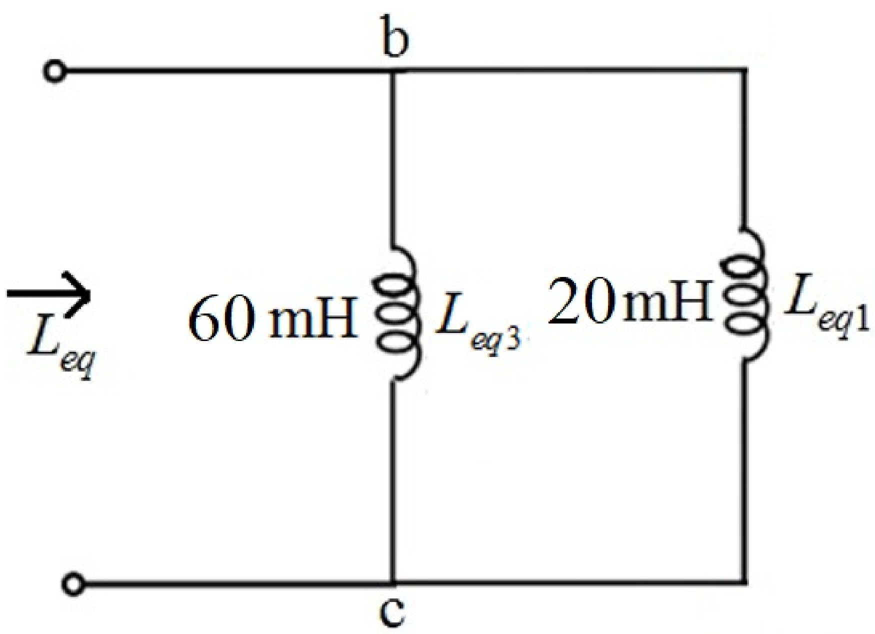


Figure 4

In Figure 4, the inductors L_{eq3} and L_{eq1} are in parallel.

$$
\begin{aligned}
L_{eq} &= \left(L_{eq3} \,||\, L_{eq1}\right) \\
&= \left(60\times10^{-3}\right) || \left(20\times10^{-3}\right) \\
&= \frac{\left(60\times10^{-3}\right)\left(20\times10^{-3}\right)}{\left(60\times10^{-3}\right)+\left(20\times10^{-3}\right)} \\
&= 15\times10^{-3} \\
L_{eq} &= 15 \text{ mH}
\end{aligned}
$$

Thus, the equivalent inductance of the circuit is $\boxed{15 \text{ mH}}$.